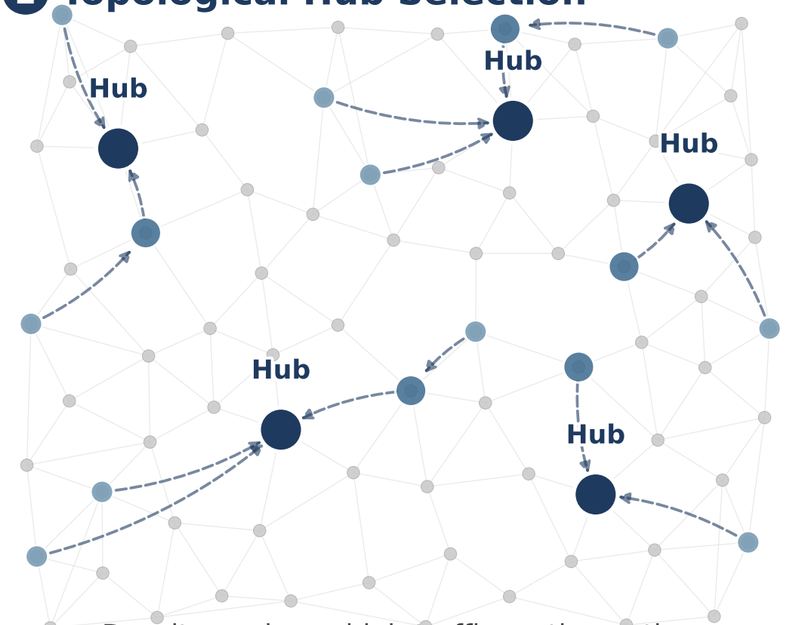
NETG: Offline Topological Distillation Pipeline

1 ACO Path Mining



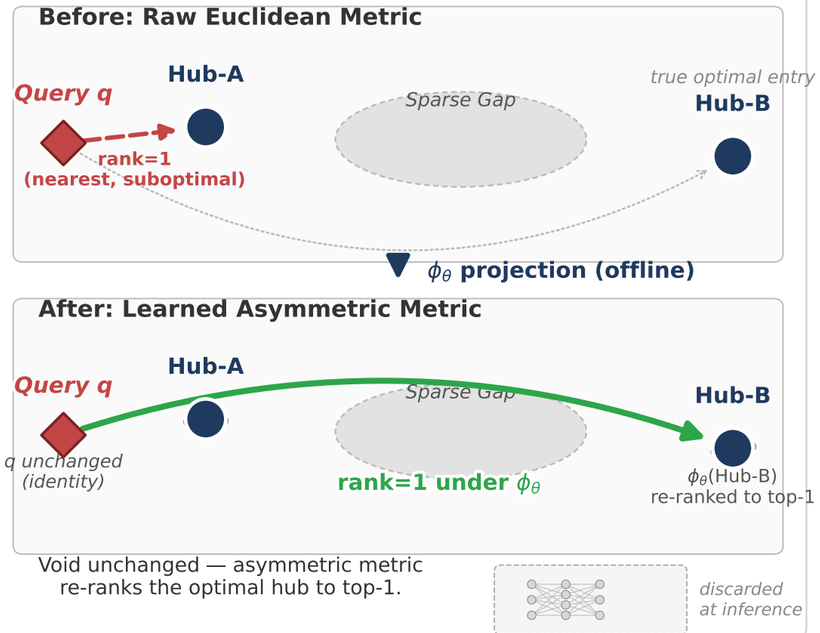
Pheromone w_i accumulates
along high-traffic routing paths.

2 Topological Hub Selection



Density peaks on high-traffic routing paths,
not geometric centroids.

3 Asymmetric Metric Learning



Hub Query Sparse Gap ACO trail

Online Serving : query (raw space) ⟶ **Micro−HNSW** on projected hubs (cache-friendly, zero neural cost) ⟶ greedy search on base graph